

Ultimate OpenCode & AI Engineering Masterguide (v6)

Expanded master repository index, core dependencies, OpenCode skills directory paths, CLI commands, and advanced workflows for HYDRA SAMO LABS.

1. Core Environment Paths & Dependencies

Arch Linux System Dependencies & OpenCode Workspace Setup

`/home/hydrasamo/Projects HYDRA/HYDRA SAMO LABS/.opencode/skills`

Essential core packages, runtimes, and shell environment utilities required for running OpenCode, Node CLI tools, and agent scripts smoothly on Arch Linux / CachyOS (Zsh & Hyprland).

How to use: Run these commands in your Zsh terminal to ensure all system dependencies are fully installed.

```
# Update system and install essential build tools, Node.js, npm, git, and python-pipx
sudo pacman -S --needed base-devel git nodejs npm python-pipx zsh

# Create the primary OpenCode skills directory structure
mkdir -p "/home/hydrasamo/Projects HYDRA/HYDRA SAMO LABS/.opencode/skills"
```

2. Design Systems & Development Frameworks

Emil Kowalski Design Skills

<https://github.com/emilkowalski/skills.git>

Provides UI micro-interactions, animation curves, and component polish guidelines.

How to use: Navigate to your skills directory and clone the repository directly.

```
cd "/home/hydrasamo/Projects HYDRA/HYDRA SAMO LABS/.opencode/skills"
git clone https://github.com/emilkowalski/skills.git emilkowalski-skills
```

UI/UX Pro Max Skill Framework

<https://github.com/nextlevelbuilder/ui-ux-pro-max-skill.git>

Advanced UI/UX architecture and prompt structure framework for high-conversion web apps.

How to use: Clone directly into your skills directory for agentic UI design reviews.

```
cd "/home/hydrasamo/Projects HYDRA/HYDRA SAMO LABS/.opencode/skills"
git clone https://github.com/nextlevelbuilder/ui-ux-pro-max-skill.git ui-ux-pro-max
```

Awesome Design MD (VoltAgent)

<https://github.com/VoltAgent/awesome-design-md>

Plain-text design tokens reverse-engineered from 55+ top developer sites (Vercel, Stripe, Linear, Raycast).

How to use: Clone into skills directory and apply brand tokens.

```
cd "/home/hydrasamo/Projects HYDRA/HYDRA SAMO LABS/.opencode/skills"
git clone https://github.com/VoltAgent/awesome-design-md.git awesome-design-md
cp awesome-design-md/design-md/vercel/DESIGN.md ../../DESIGN.md
```

3. Agent Orchestration & Specialized Subagents

ECC Evaluation & Coding Harness

<https://github.com/affaan-m/ECC.git>

Rigorous evaluation and sandbox harness designed for high-accuracy local agent execution.

How to use: Clone directly into your skills directory to enable rigorous verification suites.

```
cd "/home/hydrasamo/Projects HYDRA/HYDRA SAMO LABS/.opencode/skills"
git clone https://github.com/affaan-m/ECC.git ecc-harness
```

VoltAgent TypeScript Platform

<https://github.com/VoltAgent/voltagent>

End-to-end TypeScript framework for building memory-backed RAG agents, custom guardrails, and multi-step workflows.

How to use: Scaffold customized backend agentic servers with local database memory adapters.

```
npm create voltagent-app@latest
```

Awesome Claude Code Subagents

<https://github.com/VoltAgent/awesome-claude-code-subagents>

100+ specialized subagent personas for architecture, testing, security reviews, and devops.

How to use: Clone into skills directory to deploy specialized subagents.

```
cd "/home/hydrasamo/Projects HYDRA/HYDRA SAMO LABS/.opencode/skills"
git clone https://github.com/VoltAgent/awesome-claude-code-subagents.git claude-subagents
```

Awesome Agent Skills Index

<https://github.com/VoltAgent/awesome-agent-skills>

Curated index of 1,000+ real-world agent routines from top tech teams (Next.js, Vercel, Stripe).

How to use: Clone into skills directory to reference developer workflows.

```
cd "/home/hydrasamo/Projects HYDRA/HYDRA SAMO LABS/.opencode/skills"
git clone https://github.com/VoltAgent/awesome-agent-skills.git awesome-agent-skills
```

4. Gateways, APIs, Parsers & Video Infrastructure

OpenRouter AI Gateway & Free Tier Integration

<https://openrouter.ai/>

Unified API gateway providing access to hundreds of LLMs through a single endpoint with automated fallbacks and a virtual free router.

How to use in OpenCode: Connect via /connect inside OpenCode or configure the model target.

```
{  
  "model": "openrouter/free"  
}
```

Microsoft Markdown CLI & Python Tool

<https://github.com/microsoft/markitdown>

Universal file converter turning complex binaries (PDFs, Excel spreadsheets, audio, images) into clean Markdown.

How to use: Install via pipx and convert local documents for LLM context injection.

```
pipx install markitdown  
markitdown document.pdf > output.md
```

Google Labs Code Modules

<https://github.com/google-labs-code>

Experimental developer modules and sandbox utility scripts curated by Google Labs.

How to use: Clone into skills directory to test experimental agent capabilities.

```
cd "/home/hydrasamo/Projects HYDRA/HYDRA SAMO LABS/.opencode/skills"  
git clone https://github.com/google-labs-code.git google-labs-code
```

OmniRoute AI Gateway

<https://github.com/diegosouzapw/OmniRoute>

Universal proxy unifying 290+ providers and 500+ models into one OpenAI-compatible endpoint with token compression.

How to use: Point OpenCode/Claude Code to your local proxy port for free tier routing & fallbacks.

```
npm install -g omniroute && omniroute start
```

Freebuff Platform & CLI Agent

<https://freebuff.com/>

100% free ad-supported AI coding agent, cloud sandboxes, and deep chat alternative to paid subscriptions.

How to use: Run zero-cost parallel coding agents directly inside your terminal workspace.

```
npm install -g freebuff && cd "/home/hydrasamo/Projects HYDRA/HYDRA SAMO LABS" && freebuff
```

Remotion Video Generation Framework

<https://www.remotion.dev/> (or Remotion CLI / React)

Programmatic video creation framework allowing developers to build videos, animations, and motion graphics using React and TypeScript.

How to use: Clone or install inside your workspace/skills folder as a custom motion video skill.

```
cd "/home/hydrasamo/Projects HYDRA/HYDRA SAMO LABS/.opencode/skills"  
mkdir -p remotion-skill && cd remotion-skill  
npm install @remotion/cli @remotion/bundler remotion react react-dom
```

Video Links & Titles

- **Video 1:** [Building Modern 3D Web Experiences](#) (*Note: Part of a 3D web development playlist*)
- **Video 2 & 6:** [Advanced Motion Graphics and Code Setup](#)
- **Video 3:** [AI Agent Workflow Integration Tutorial](#)
- **Video 4:** [Hyprrland Window Manager Customization Guide](#)
- **Video 5 (Must See):** [Mastering Remotion for Programmatic Video Creation](#)
- **Video 7:** [Zsh and Shell Scripting Optimization Tips](#)
- **Video 8:** [Full-Stack AI Framework Setup and Architecture](#)